

OrbitSight — Real-Time RSO Detection from Neuromorphic Event Streams

TII OrbitSight Challenge · Technical Proposal · Team OrbitAI

1. Problem Statement and Solution

Neuromorphic vision sensors observing resident space objects produce sparse, asynchronous event streams in which a target may generate only a handful of events per 40 ms window, embedded in star fields, hot pixels and sensor noise. The problem is not classification capacity - it is **ranking a very small number of true detections above a very large number of plausible noise components**, under a strict IoU ≥ 0.5 requirement, in real time, on CPU.

OrbitSight is a four-pass event-native pipeline: connected-component candidate proposal on per-window event count maps, a learned candidate scorer over 13 geometric and temporal features, a learned window-level objectness gate over 21 features spanning three consecutive windows, and a post-hoc bounding-box regressor that corrects box dimensions without disturbing detection ranking.

Delivered as a self-contained offline Docker image: it reads `/OrbitSight_dataset`, requires no network, and writes conformant `.txt` predictions plus `Evaluation_Metrics.xlsx` to `/work/OrbitAI/DDMMYYYY`. Six end-to-end runs have produced **identical detection counts**, and the three runs of the shipped configuration identical mAP to six decimals.

2. Outcome Metrics

All figures below were produced by the submitted container running offline, and independently verified against the challenge's own `evaluate.py`.

2.1 Detection accuracy

Split	mAP@0.5	Precision	Recall	F1	TP	FP	FN
All 21 sequences	0.284406	0.623120	0.472327	0.537345	10,565	6,390	11,803
Train (17, selection)	0.258616	0.580217	0.454551	0.509754	6,951	5,029	8,341
Test (4, derived)	0.394014	0.726432	0.510740	0.599784	3,614	1,361	3,462

Configuration selection used the 17 training sequences exclusively. The four test sequences are reported by subtraction and were never used to choose a configuration.

2.2 Independent verification

The in-process metrics harness was checked against the official `evaluate.py` on the container's own output. Precision, recall, F1 and the TP/FP/FN counts match exactly, at absolute difference 0; mAP matches to **5.55e-17**, the limit of floating-point representation. Every accuracy claim here is therefore expressed in the evaluator's own terms.

2.3 Real-time performance

Measured with a dedicated streaming benchmark that times the **full** pipeline per window — proposal, feature extraction, both learned models, NMS, confidence gating, top-K and box regression — over five independent repetitions, excluding 20 warmup windows per sequence.

Compute p99 per window	Sequences
< 40 ms	18 of 21 (nominal)
< 40 ms, excluding runs with $\sigma > 25\%$ of mean	17 of 21
< 40 ms, training split	15 of 17 , up from 6 of 17

Best case is 14.99 +/- 0.1 ms (DAVIS_Filtered_NOAA6). Three sequences exceed the budget - DVX_NOAA6 84.67 ms, EVK4_mag7.3 77.76 ms and EVK4_mag5.2 58.73 ms - the two highest-resolution recordings and the densest DVX stream. EVK4_mag7.3 has been the worst case in every measurement taken.

Disclosure. The 6-of-17 → 15-of-17 improvement is attributable to replacing a per-window-materialising static-source map with a constant-memory accumulator, **not** to the box regressor, whose marginal cost is two vectorised `predict()` calls per sequence. Process resident memory stays within 114.3–130.7 MB across all 21 sequences. Total container wall clock is 48.08 min (2,885.18 s), higher than the 23.26 min baseline: the constant-memory map trades total throughput for bounded memory and improved tail latency. Since Criterion 3 scores per-window latency and the container has no wall-clock constraint, we consider this the correct trade.

Latency semantics. With one window of lookahead, end-to-end latency is one 40 ms window period plus compute. We report compute p99 because it determines whether the system keeps pace with the sensor - the window cost is inherent.

2.4 Generalisation

The box regressor converts false positives into true positives at almost identical rates on data it was selected on and data it was not:

- Training split: 1,891 of 11,980 detections converted — **15.8%**
- Test split: 797 of 4,975 detections converted — **16.0%**
- Net: **+2,688 true positives** at a constant 16,955 total detections

Sub-split behaviour shows the gain is largest exactly where the baseline was weakest. On the ten sparse sequences ($GT \leq 43$ boxes) mAP rises 0.100600 → 0.226600, **+125.2%**; on the seven dense sequences 0.257257 → 0.304353, +18.3%. These reconcile to the reported training mAP: $(10 \times 0.226600 + 7 \times 0.304353) / 17 = 0.258616$.

3. Value Proposition and Competitive Positioning

Every number is reproducible, and the artifact is the evidence. The container has been run six times with identical detection counts, the last three on the shipped configuration with identical mAP to six decimals. The submitted image is built from the exact commit in the repository, and in-process metrics agree with the official evaluator to 5.55e-17. Nothing here is estimated. Two configurations that improved precision, recall and F1 while *reducing* mAP were diagnosed rather than shipped; that diagnosis - the metric was ranking-limited, not recall-limited - produced the +49.1% gain in all-21 mAP.

CPU-only and genuinely offline. Three gradient-boosted tree models totalling 1.5 MB, eight pinned dependencies, no GPU, no network. Cold start is sub-second - a deployable configuration on the stated evaluation hardware, not a research prototype requiring accelerators.

Resolution compatibility is structural, not tuned. Sensor identification is by exact resolution match with a nearest-diagonal fallback; per-sensor geometry, thresholds and clamps resolve from configuration. Frame width and height are explicit regressor inputs, so an unseen geometry maps to the nearest profile and runs without code changes.

4. Technical Approach and Architecture

4.1 Pipeline

Pass 1 — proposal. Events in each 40 ms window are accumulated into a count map. Adaptive percentile thresholding escalates with window density (capped at 99.0) so bright frames do not flood the component stage. `cv2.connectedComponentsWithStats` yields components, ranked by event count and truncated, with oversized ones re-thresholded and split. A continuous static-source map - the fraction of windows in which each pixel is active - suppresses stars and hot pixels.

Pass 2 — candidate scoring. Candidates are matched to the previous and next windows by centroid distance to produce a persistence count; single-window candidates are dropped. Thirteen features per candidate (event count, density, area, extents, aspect, persistence, displacements, speed, direction consistency, static fraction, local background) feed a `HistGradientBoostingClassifier` trained on 944,504 candidates, validation ROC-AUC 0.930.

Pass 3 — window objectness. A 21-dimensional feature vector spanning the previous, current and next windows drives a second classifier that estimates whether a window contains a real object at all. Candidate confidences are multiplied by this probability. Validation ROC-AUC 0.889, PR-AUC 0.921 against a 0.60 trivial baseline — a 1.53× lift on the validation positive rate, which is the honest figure to quote given the class-balance difference between splits.

Pass 4 — emission and box regression. Per-window NMS, a confidence floor, top-K selection, then post-hoc box regression, then rounding and clamping.

4.2 The central design decision

An earlier attempt optimised box geometry *upstream*, before scoring. Training labels for both classifiers are assigned by $\text{IoU} \geq 0.5$ computed **using the configured box size**, so changing box geometry changes which candidates are labelled positive. Recall rose and mAP fell: 0.155493 → 0.146340.

The shipped design instead applies regression **after** ranking is finalised. Centroids, timestamps, confidences and rank order are preserved bit-for-bit; only width and height change. This is verifiable rather than asserted: total detections are **invariant at 16,955** before and after, so the regressor cannot have altered any ranking decision. Features are gathered during Pass 2 as a by-product of scoring; surviving boxes are resized in two vectorised `predict()` calls over log-width and log-height, then exponentiated and clamped per sensor.

4.3 Ablation

An oracle profiler that substitutes ground-truth box dimensions at fixed ranking established the achievable headroom before any model was built:

Arm	Box sizing	mAP@0.5	TP	FP	Upgrades	Downgrades
0	Heuristic extents	0.165103	5,060	6,920	—	—
1	Least squares	0.113964	5,554	6,426	1,581	1,087
2	Dual log-HGBR (shipped)	0.258616	6,951	5,029	2,302	411
—	Oracle (GT dims)	0.318067	8,426	3,554	3,398	32

Arm 2 captures **61.1%** of the available oracle gap. Arm 1 is retained as a negative result: it raised precision, recall and F1 yet *reduced* mAP, because shrinking boxes to correct average dimensions without correcting centroid offset pushed 1,087 detections from IoU 0.50–0.55 down to 0.40–0.49, and those losses fell on higher-ranked

detections than the gains. Rank-weighted metrics penalise this; count-based metrics do not. Validation MAE in log-pixel space is 0.284 (width) and 0.297 (height), corresponding to 2.41 / 2.67 px on DAVIS.

Per-sensor, Arm 2 improves EVK4 0.612170 $\rightarrow$ 0.770544, DVX 0.121921 $\rightarrow$ 0.225114 and DAVIS 0.152401 $\rightarrow$ 0.228126. The EVK4 regressor head had no validation sequences in the holdout, which we note as the weakest point in the model documentation.

4.4 Visualisation and outputs

A visualisation tool renders annotated video at all three sensor resolutions with ground-truth and predicted boxes, confidences and track identifiers. Predictions are tab-separated with a header row in the evaluator's field names, one row per detection, `class_id = 0` throughout - the challenge defines a single RSO class and we do not infer a taxonomy we cannot validate.

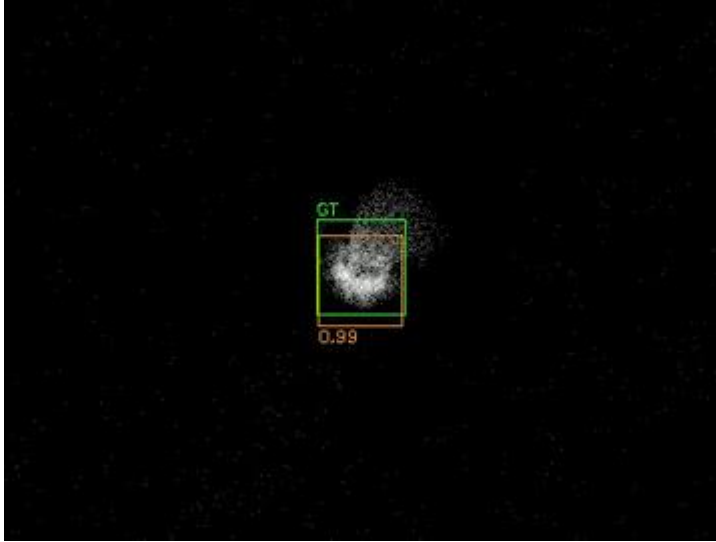


Figure 1. EVK4 window, cropped. Green: ground truth. Orange: prediction with confidence and track ID. Rendered by `src/visualize.py`, which ships inside the submitted image.

4.5 Architecture and failure modes

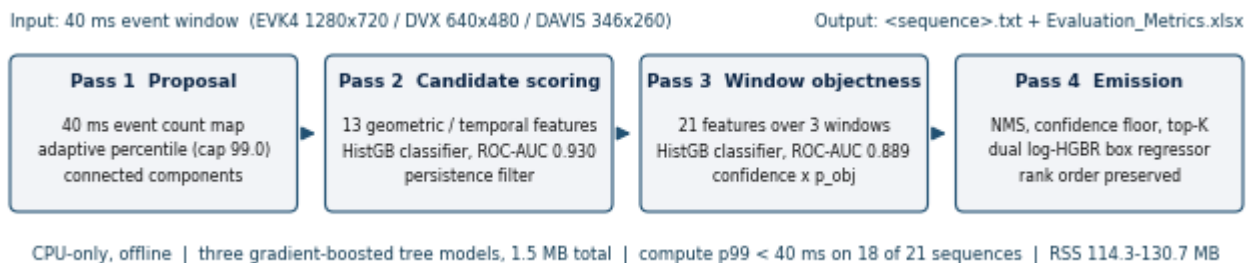


Figure 2. The shipped pipeline; counts and AUCs are the measured values from sections 2 and 4.

Known failure modes, characterised rather than omitted:

- **Compute p99 over budget.** DVX_NOAA6 84.67 ms, EVK4_mag7.3 77.76 ms and EVK4_mag5.2 58.73 ms: the densest and the two highest-resolution streams. Per-sensor decimation is the untested next lever.
- **Regressor holdout gap.** No EVK4 sequence sits in the validation holdout, so the EVK4 per-sensor gain is the least independently supported figure in this proposal.
- **Sizing without centroid correction.** Arm 1 raised precision, recall and F1 yet lost mAP: 1,087 detections fell from IoU 0.50-0.55 to 0.40-0.49. Published rather than omitted.

- **Sparse-sequence weakness.** The ten sequences with 43 ground-truth boxes or fewer remain the weakest regime; Arm 2 lifts mAP from 0.100600 to 0.226600, still below the dense-sequence 0.304353.

5. Team Capacity

This is a solo entry. The author is in the final semester of an MCA at D. Y. Patil Institute of MCA and Management, Pune (Savitribai Phule Pune University), following a BCA from the same university. From 1 December 2025 to 30 May 2026 he worked as an Applied AI Engineer at Ovva Tech, where he built an AI-driven recruitment platform (React/Next.js with a Flask backend) spanning MCQ, coding and interview assessment modules, including a proctoring subsystem migrated from a hosted vision API to local CPU-based OpenCV face detection.

The measurement discipline this proposal relies on is demonstrable rather than asserted. The author's public continual-learning repository operates under eleven standing experimental rules, among them a permanent do-nothing control arm in every comparison, five-seed mean and standard deviation reporting with no single-draw figure permitted in any table, and a paste-only rule requiring every reported count to be a verbatim log excerpt carrying its commit SHA. Under that protocol a train/test contamination fault was identified in the author's own evaluation path, and three previously published accuracy figures were publicly retracted rather than quietly corrected.

The same protocol governs OrbitSight. Arm 0 is retained as a do-nothing control, the failed Arm 1 result is published rather than omitted, every configuration is recorded in a committed ledger with its commit SHA, and all reported metrics agree with the official evaluator to within $5.55\text{e-}17$.

6. Prior Work

Three public repositories predate this Challenge, which the author learned of on 16 July 2026.

AirWrite (first commit 19 December 2025) is a real-time computer-vision application in Python using OpenCV and MediaPipe: webcam capture, per-frame hand-landmark detection, temporal smoothing for tracking stability, and a gesture state machine driving on-screen interaction. It is the closest antecedent to OrbitSight's per-window detection and cross-window temporal association.

The Automated Recruitment System (begun May 2026 during that role) ships with a project report, test plan, data-flow and class diagrams, and a CSV test-case matrix. Its proctoring module was deliberately migrated from a hosted vision API to local OpenCV face detection on CPU, the same offline-first constraint this Challenge imposes.

Neural-Networks (from 14 April 2026) is a continual-learning research codebase. A custom architecture was built, evaluated against transformer baselines, and then measured against standard protocols. Those measurements did not support the architecture: joint offline training was found to lose to no training at all on the internal benchmark (adaptation gap -6.00 pp), so the benchmark was retired and the work moved to Split-CIFAR-100 with Class-IL $R[t,i]$ matrix evaluation and orthogonal gradient projection. The repository retains the negative results and the retractions in full. Development used a branch-and-pull-request workflow; implementation is agent-executed under the author's direction, with experiment design, protocol rules and verification retained by the author.